KKS-KHS Formulation

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1 Model Description

Phase field model is being used for shrinking of annulus, where thermodynamics, kinetics and mechanics are coupled together. Detailed description of each of the parts are described below.

1.1 Thermodyanmics

We have two phase (fuel, annulus) and three component (c_{zr}, c_v, c_u) system. The corresponding chemical free energy density for each of the two phase system is described based on the approximated parabolic forms for free energies.

$$F = \int_V \left(f_{\text{bulk}} + f_{\text{int}} + f_{\text{el}} \right) dV. \tag{1}$$

$$f_{\text{bulk}} = f_{\text{bulk}}^{\text{fuel}}(1-h) + f_{\text{bulk}}^{\text{annulus}} h. \tag{2}$$

$$\begin{aligned} f_{\text{bulk}}^{\text{fuel}} = \; & \left(0.225 - c_{\text{Zr,fuel}}\right)^2 + 30\left(10^{-6} - c_{\text{v,fuel}}\right)^2 \\ & + \left(1 - c_{\text{Zr,fuel}} - c_{\text{v,fuel}} - 0.775\right)^2. \end{aligned} \tag{3}$$

$$\begin{aligned} f_{\text{bulk}}^{\text{annulus}} = \; & 100\left(0 - c_{\text{Zr,annulus}}\right)^2 + 30\left(1 - c_{\text{v,annulus}}\right)^2 \\ & + 30\left(1 - c_{\text{Zr,annulus}} - c_{\text{v,annulus}} - 0\right)^2. \end{aligned} \tag{4}$$

$$f_{\text{int}} = W g(\eta) + \frac{\kappa}{2} \left|\nabla \eta\right|^2, \tag{5}$$

$$g(\eta) = \eta^2 (1-\eta)^2. \tag{6}$$

$$\kappa = \frac{6 l \sigma}{\alpha}, \qquad \alpha = 2.2. \tag{7}$$

$$W = \frac{3\alpha\sigma}{l}, \qquad \alpha = 2.2. \tag{8}$$

$$h_{\text{fuel}}(\eta) = 1 - h(\eta) \tag{9}$$

$$h(\eta) = \eta^3 \left(6\eta^2 - 15\eta + 10\right) \tag{10}$$

1.2 Kinetics

For kinetics of the model, since we have two phases fuel and annulus, we define the corresponding diffusivities for Zirconium and Vacancy based on the literature values. In this model, we are assuming that the fuel (solid) is entirely composed of gamma phase U-Zr, therefore the kinetics data used in this model is based on the gamma kinetics of U-Zr alloys. Since the vacancy diffusivity data for gamma U-Zr phase is not available, we will make it a function of Zr diffusivity and choose a significantly higher value for it, as the diffusion process is Zr limited. Below, we have described the equations for Zr kinetics as obtained from literature.

$$M_{\text{Zr,eff}} = \frac{D_{\text{Zr,eff}}}{\dfrac{\partial^2 f_{\text{fuel}}}{\partial c^2_{\text{Zr,fuel}}}} \tag{11}$$

$$D_{\text{Zr,eff}} = D_{\text{Zr}} h_{\text{fuel}} + 0.5 D_{\text{Zr}} \left(1 - h_{\text{fuel}}\right) \tag{12}$$

$$D_{\text{Zr}} = D_0 \exp\left(-\frac{Q}{RT}\right) \tag{13}$$

$$D_0 = \exp\left(-6.15 - 18.5 c_{\text{Zr,fuel}} + 21 c^2_{\text{Zr,fuel}}\right) \tag{14}$$

$$Q = \left(128 - 107 c_{\text{Zr,fuel}} + 174 c^2_{\text{Zr,fuel}}\right) \tag{15}$$

Vacancy diffusivity are described below,

$$D_{\text{v,eff}} = 100 D_{\text{Zr}} h_{\text{fuel}} + 200 D_{\text{Zr}} \left(1 - h_{\text{fuel}}\right) \tag{16}$$

$$M_{\text{vac,eff}} = \frac{D_{\text{v,eff}}}{\dfrac{\partial^2 f_{\text{fuel}}}{\partial c^2_{\text{v,fuel}}}} \tag{17}$$

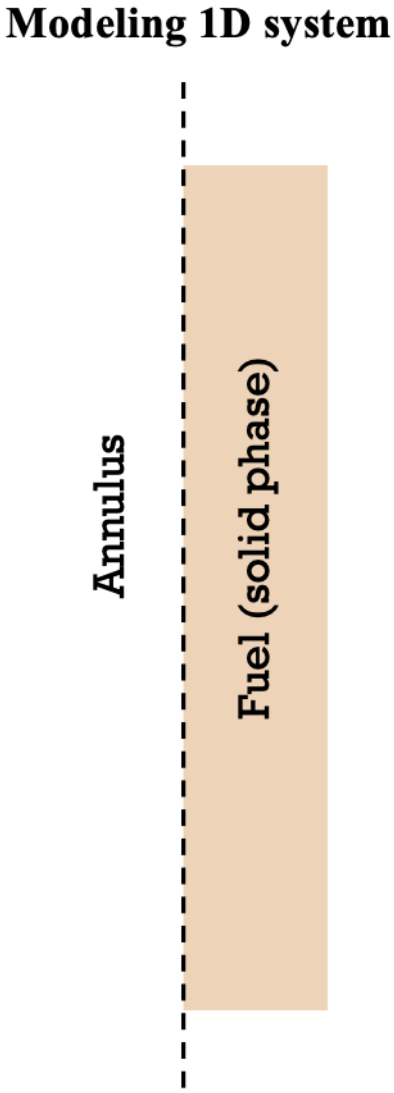


Figure 1: Simulation of fuel and annulus in RZAxisymmetric System

1.3 Mechanics

For mechanics, we are making use KHS scheme of interpolation, for coupling mechanics model with KKS phase field model. In KHS, basic idea is that we define the mechanical properties such as elastic stiffness constant, eigen strains to be interpolated for the given phases. Then based on these properties, we calculate global stress, global elastic energy which is then coupled to KKS. Mathematical flow of approach is described below,

$$C_{ijkl} = C_{ijkl}^{\text{fuel}} h_{\text{fuel}} + C_{ijkl}^{\text{annulus}} h \tag{18}$$

$$\varepsilon_{ij} = \varepsilon_{\text{total}} - \varepsilon_{\text{Zr}}^{*} h_{\text{fuel}} - \varepsilon_{\text{vac}}^{*} h_{\text{fuel}} - \varepsilon_{\text{thermal}}^{*} h_{\text{fuel}} \tag{19}$$

$$\boldsymbol{\varepsilon}_{\text{Zr}}^{*} = 0.05 \left(c_{\text{Zr,fuel}} - 0.225\right) \boldsymbol{I} \tag{20}$$

$$\boldsymbol{\varepsilon}_{\text{vac}}^{*} = 0.05 \left(c_{\text{vac,fuel}} - 10^{-6}\right) \boldsymbol{I} \tag{21}$$

$$\frac{\Delta L}{L_0} = 1.0003 \times 10^{-2} + 7.43 \times 10^{-5} \left(T - 868\right) \tag{22}$$

$$\boldsymbol{\varepsilon}_{\text{thermal}}^{*} = \alpha_T \left(T - T_0\right) \boldsymbol{I} \tag{23}$$

substituting the value of eq (18) (19) and (21) in eq (17)

$$\varepsilon_{ij} = \varepsilon_{ij}^{\text{total}} - \left[0.05 \left(c_{\text{Zr,fuel}} - 0.225\right) + 0.05 \left(c_{\text{vac,fuel}} - 10^{-6}\right) + \alpha_T \left(T - T_0\right)\right] I_{ij}\, h_{\text{fuel}} \tag{24}$$

Based on the above parameters, we calculate the global stress and global elastic energy.

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl} \tag{25}$$

$$f_{\text{elastic}} = \frac{1}{2} C_{ijkl} \varepsilon_{ij} \varepsilon_{kl} \tag{26}$$

Now the elastic energy is added to the fuel phase chemical free energy density and not to annulus since there is not eigen strains in annulus. total free energy densities including chemical and elastic parts can be written as,

$$f_{\text{total}}^{\text{fuel}} = f_{\text{bulk}}^{\text{fuel}} + f_{\text{elastic}} \tag{27}$$

$$f_{\text{total}}^{\text{annulus}} = f_{\text{bulk}}^{\text{annulus}} \tag{28}$$

1.4 KKS Coupling Mechanics

Here we couple the total free energy densities with KKS equations.

KKS Phase Concentration Constraints.

$$c_{\mathrm{Zr}} = (1-h)\, c_{\mathrm{Zr,fuel}} + h\, c_{\mathrm{Zr,annulus}} \tag{29}$$

$$c_{\mathrm{v}} = (1-h)\, c_{\mathrm{v,fuel}} + h\, c_{\mathrm{v,annulus}} \tag{30}$$

Pointwise equality

$$\mu^{\mathrm{fuel}} = \mu^{\mathrm{annulus}} \tag{31}$$

$$\frac{\partial f_{\mathrm{chem}}^{\mathrm{fuel}}}{\partial c_{\mathrm{Zr,fuel}}} + \frac{\partial f_{\mathrm{elastic}}}{\partial c_{\mathrm{Zr,fuel}}} = \frac{\partial f_{\mathrm{chem}}^{\mathrm{annulus}}}{\partial c_{\mathrm{Zr,annulus}}} \tag{32}$$

$$\frac{\partial f_{\mathrm{chem}}^{\mathrm{fuel}}}{\partial c_{\mathrm{v,fuel}}} + \frac{\partial f_{\mathrm{elastic}}}{\partial c_{\mathrm{v,fuel}}} = \frac{\partial f_{\mathrm{chem}}^{\mathrm{annulus}}}{\partial c_{\mathrm{v,annulus}}} \tag{33}$$

Total Free Energy in KKS coupled mechanics model:

$$F = \int_V \left[(1-h)\left(f_{\mathrm{fuel}}\right) + f_{\mathrm{elastic}} + h f_{\mathrm{annulus}} + W g(\eta) + \frac{\kappa}{2} \left|\nabla \eta\right|^2 \right] dV \tag{34}$$

The Residuals for the governing equations implemented in this model with coupled elasticity:

$$\frac{\partial \eta}{\partial t} = -L \frac{\delta F}{\delta \eta} \tag{35}$$

There are 4 residuals for allen cahn equation implemented by KKS allen cahn kernels and one additional residual comes fro elastic free energy coupling.

$$R_\eta^{\mathrm{Time}} = \int_\Omega \psi \, \frac{\partial \eta}{\partial t} \, \mathrm{d}\Omega \tag{36}$$

$$R_\eta^{\mathrm{KKSACBulkC,Zr}} = \int_\Omega \psi L h'(\eta)\, w_1 \left(c_{\mathrm{Zr}}^{\mathrm{annulus}} - c_{\mathrm{Zr}}^{\mathrm{fuel}}\right) \, \mathrm{d}\Omega \tag{37}$$

$$R_\eta^{\mathrm{KKSACBulkC,v}} = \int_\Omega \psi L h'(\eta)\, w_2 \left(c_{\mathrm{v}}^{\mathrm{annulus}} - c_{\mathrm{v}}^{\mathrm{fuel}}\right) \, \mathrm{d}\Omega \tag{38}$$

$$R_\eta^{\mathrm{KKSACBulkF}} = \int_\Omega \psi L \left[h'(\eta) \left(f^{\mathrm{fuel}} - f^{\mathrm{annulus}}\right) + W g'(\eta) \right] \, \mathrm{d}\Omega \tag{39}$$

$$R_\eta^{\mathrm{Elastic}} = \int_\Omega \psi L \frac{\partial f_{\mathrm{el}}}{\partial \eta} \, \mathrm{d}\Omega \tag{40}$$

$$R_\eta^{\mathrm{ACInterface}} = \int_\Omega L \kappa \nabla \eta \cdot \nabla \psi \, \mathrm{d}\Omega \tag{41}$$

Next governing equation we have is Modified Cahn-Hilliard Equation with soret term:

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[M_c \nabla \left(\frac{\delta F}{\delta c} \right) - M_T \frac{\nabla T}{T} \right] \tag{42}$$

KKSSplitCHCRes:

$$\frac{\partial f_{\text{chem}}^{\text{fuel}}}{\partial c_{\text{Zr,fuel}}} + \frac{\partial f_{\text{elastic}}}{\partial c_{\text{Zr,fuel}}} - w_1 = 0 \tag{43}$$

$$\frac{\partial f_{\text{chem}}^{\text{annulus}}}{\partial c_{\text{Zr,annulus}}} - w_1 = 0 \tag{44}$$

$$\frac{\partial f_{\text{chem}}^{\text{fuel}}}{\partial c_{\text{v,fuel}}} + \frac{\partial f_{\text{elastic}}}{\partial c_{\text{v,fuel}}} - w_2 = 0 \tag{45}$$

$$\frac{\partial f_{\text{chem}}^{\text{annulus}}}{\partial c_{\text{v,annulus}}} - w_2 = 0 \tag{46}$$

SplitCHWRes:

$$R_{\text{Zr}}^{\text{SplitCHWRes}} = \int_{\Omega} M_{\text{Zr,eff}} \nabla w_1 \cdot \nabla \psi_{\text{Zr}} \, d\Omega \tag{47}$$

$$R_{\text{v}}^{\text{SplitCHWRes}} = \int_{\Omega} M_{\text{vac,eff}} \nabla w_2 \cdot \nabla \psi_{\text{v}} \, d\Omega \tag{48}$$